Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional

Final Report

Project Title : Farm Droid
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Contents

Farm Droid

An Autonomous Agricultural Robot

1. Introduction

Labour shortage and the requirement to monitor crop health with precision are huge problems in agriculture. To tackle these challenges, it has developed a novel agricultural robot concept known as “Farm Droid”. To overcome these challenges, it has come up with a novel idea for an agricultural robot, dubbed “Farm Droid”, which is compact and autonomous.

Farm Droid is the low-cost, adaptable mobile platform that can be used in place of traditional heavy machinery. This enables it to maneuver between rows of crops efficiently to carry out monitoring and manipulating. It combines cutting-edge embedded technologies: a Raspberry Pi 4 computer for computer vision and decision making, and an arduino UNO/ nano board for sensor data collection and real time motor control.

2. Objectives

The main goal is to create a smart farming assistant that will be able to work autonomously. The aims are:

- **Realtime Monitoring:** To include sensor measuring soil moisture, pH and weather conditions (Rain/Temperature).
- **Modular Design:** Servo Driven Arm Mechanism for Various Modular Tasks: Robotic Manipulation, Weeding.
- **Edge Computing:** To use a Raspberry Pi Camera with OpenCV to identify crops, weeds and obstacles in real-time.
- **Seamless Connectivity:** To enable a reliable remote monitoring and control via the telegram bot integration or a custom mobile application.
- **Adaptive Mobility:** To design an effective locomotion system (Rover or Legged) to operate on uneven agricultural surface.
- **Autonomous Activity:** To make the robot do any other activities such as seeding of plants, watering plants, fertilizing soil, etc., without any human intervention.

3. Problem Statement

Traditional large-scale agricultural equipment may not be suitable for small-scale precision agriculture or sensitive crop fields. Large tractors compact soil do not have the agility to do individual plant inspection. The robotic systems available today are either expensive or complex enough to be too large to operate in these environments. This is where Farm Droid comes in with its lightweight, intelligent and versatile solution.

4. Scope

The project scope also includes the mechanical assembly of the robot and programming of the circuit, as well as the integration of the circuit into the robot. The hardware design consists of a modified chassis, motor / servo drive system, and a power distribution system to insure stable operation of the hardware components. The apparatus is basically made of a camera and a motor.The software side of the system includes OpenCV vision processing using Python and motor control based on a C++ firmware. The challenge with this project is that this prototype won't be deployed at a large scale industrial application level; it will only be spun in a small scale environment such as a garden or a test bed under control.

5. Methodology

5.1. System Architecture

The computational load of the system is efficiently dealt with with a "Distributed Computing" approach. It is an architecture that includes the Brain (Raspberry Pi 4), responsible on the high level for the decision making, image processing (OpenCV) and path planning. It also communicates with other modules with interfaces such as Serial, I2C and USB. The Controller (Arduino Uno / Nano) is used for low level real time control, for instance sending PWM impulses to motor control drivers or reading analog sensor data.

5.2. Technology Stack

- **Language**: Python (Vision/Logic), C++ (Arduino Firmware)
- **Accelerator**: Intel® RealSense™ depth camera (Optional for object detection)
- **Hardware Interface:** I2C (Inter-Integrated Circuit), USB.
- **Power Management:** 65W/22W 30000mAh (Buck Converters) for system stability.

6. Hardware Components

The Farm Droid consists of the following elements:

- **Actuators:** DC Gear Motors or MG996R Servos (Mobility), SG90 (Gripper/Arm), GA25 (Metal Gear motor).
- **Sensors:** Ultrasonic (HC-SR04), Soil Moisture, Rain Sensor, DHT11, IR (Infrared), LDR (Light Dependent Resistor), DFPlayer Mini, RFID Reader(Radio Frequency Identification), Relay, PH Electrode, Gas sensor(MQ135).
- **Controllers:** Raspberry Pi (4 model B (4Gb)), arduino Uno / Nano.
- **Drivers:** L298N/Motor Driver (for Wheels).
- **Vision:** Official Raspberry Pi Camera Module with Pi camera 5MP.
- **Power Supplier:** LM2596 Buck Converter, 65W/22W 30000mAh Power Bank.

7. Budget Estimation

The costs of the various components that will be needed for the "Farm Droid" are summarized below and based on actual market prices in Bangladesh. .

Category	Component Name	Unit (BDT)	Qty	Total (BDT)
Controllers	Raspberry Pi 4 Model B (4GB) [6]	14,100	1	14,100
	Arduino Uno R3 [4]	988	2	1,976
Motors & Drivers	GA25 Metal Gear Motor [3]	450	2	900
	DSServo DS3218 20KG Digital Metal Servo [4]	2350	2	4700
	Servo Motor MG996R [6]	500	1	500
	Servo Motor SG90 (Micro) [4]	180	2	360
Sensors	HC-SR04 Ultrasonic Sensor [2]	99	1	99
	FC-51 IR Obstacle Sensor [2]	45	3	135
	HC-SR501 PIR Motion Sensor [4]	93	1	93
	OV7670 Camera Module [2]	400	1	400
	DHT11 [3]	95	1	95
	LDR (Light Dependent Resistor) [3]	5	1	5
	DFplayer [4]	350	1	350
	PIR Sensor [4]	93	1	93
	RFID [4]	190	1	190
	Relay [4]	80	4	320
	PH Sensor [4]	2584	1	2584
	Moisturizer Sensor [4]	70	1	70
	Rain Sensor [4]	78	1	78
	MQ135 [4]	135	1	135
Power & Audio	Breadboard Power Supply [4]	76	1	76
	3 Watt 8 Ohm Mini Speaker [4]	250	1	250
	Electret Microphone [6]	15	2	30
Mechanical & Misc	PVC Pipe	300	5	1,500
	Pipe (Small)	30	5	150
	Half-Size Breadboard [2]	75	2	150
	Jumper Wires (Male-Male) [2]	100	4	400
	USB Cable (Arduino) [4]	150	2	300
	Screws / Hardware	100	10	1,000
	Copper Wire (26 SWG) [2]	110	2	220
Tools	Digital Scale (10kg) [1]	259	1	259
	Hex Saw	15	2	30
	Sand Paper	30	3	90
			Grand Total Estimated Cost	**31488**

Table 1: Detailed Budget Estimation

8. Analysis of cost-benefit

This section explains the cost-benefit analysis of the "Farm Droid" from a financial feasibility/ practical viability viewpoint . This analysis contrasts the initial development and operational costs against the tangible and intangible benefits generated over a projected 3-year operational lifecycle.

8.1. Cost breakdown

The costs can be broken down to initial capital spending (CapEx) and regular operating spending (OpEx).

Cost Category	Amount (BDT)	Amount (USD)
Capital Expenditure (CapEx)	33,000	270
Operational Expenditure (OpEx - Annual)	3,000	25
Total anticipated cost over three Years		42,000

Table 2: Cost breakdown and total anticipated cost over three Years

8.2. Benefit analysis

Benefit Type	Description
Tangible Benefits	Automated labor savings, yield protection through early detection, and resource efficiency lead to an estimated annual financial benefit of BDT 27,000.
Intangible Benefits	Soil preservation through reduced compaction, data-driven insights for better crop management, and improved safety for farm workers.

Table 3: Summary of Tangible and Intangible Benefits

8.3. Market Comparison: Farm Droid vs. Commercial Alternatives

The overall estimated budget is about 258 USD (with an exchange rate of 122 BDT/USD). Comparing with commercial robots, we get the following insights:

Robot Model	Brand	Primary Function	Price (USD)
Farm Droid (Our Prototype)	**Custom**	**Precision Monitoring & Manipulation**	**258**
ATEAGO 50W [7]	ATEAGO	Remote Lawn & Terrain Maintenance	4,798
ATEAGO 50PRO [8]	ATEAGO	Remote Lawn & Terrain Maintenance	7,199
ATEAGO 80PRO [9]	ATEAGO	Heavy Terrain Maintenance	14,999
ADIR Basic [10]	Ant Robotics	Autonomous Ag-Logistics & Scouting	23,292
ADIR Power [11]	Ant Robotics	Heavy Autonomous Ag-Logistics	29,115

Table 4: Price Comparison with Commercial Agriculture Robots (Source: RobotShop)

9. Work Timeline

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Analysis & Planning	✓					
Mechanical Assembly (Chassis & Drive)	✓	✓				
Circuit Integration & Power Distribution		✓	✓			
Sensor Installation & Wiring		✓	✓			
Motor Control & Calibration			✓	✓		
Arm Control & Actuation Testing				✓	✓	
Software Integration				✓	✓	✓
System Testing & Debugging					✓	✓
Final Review & Documentation						✓

Table 5: Gantt Chart for Farm Droid Development

10. Visual Models

10.1. Circuit Diagram

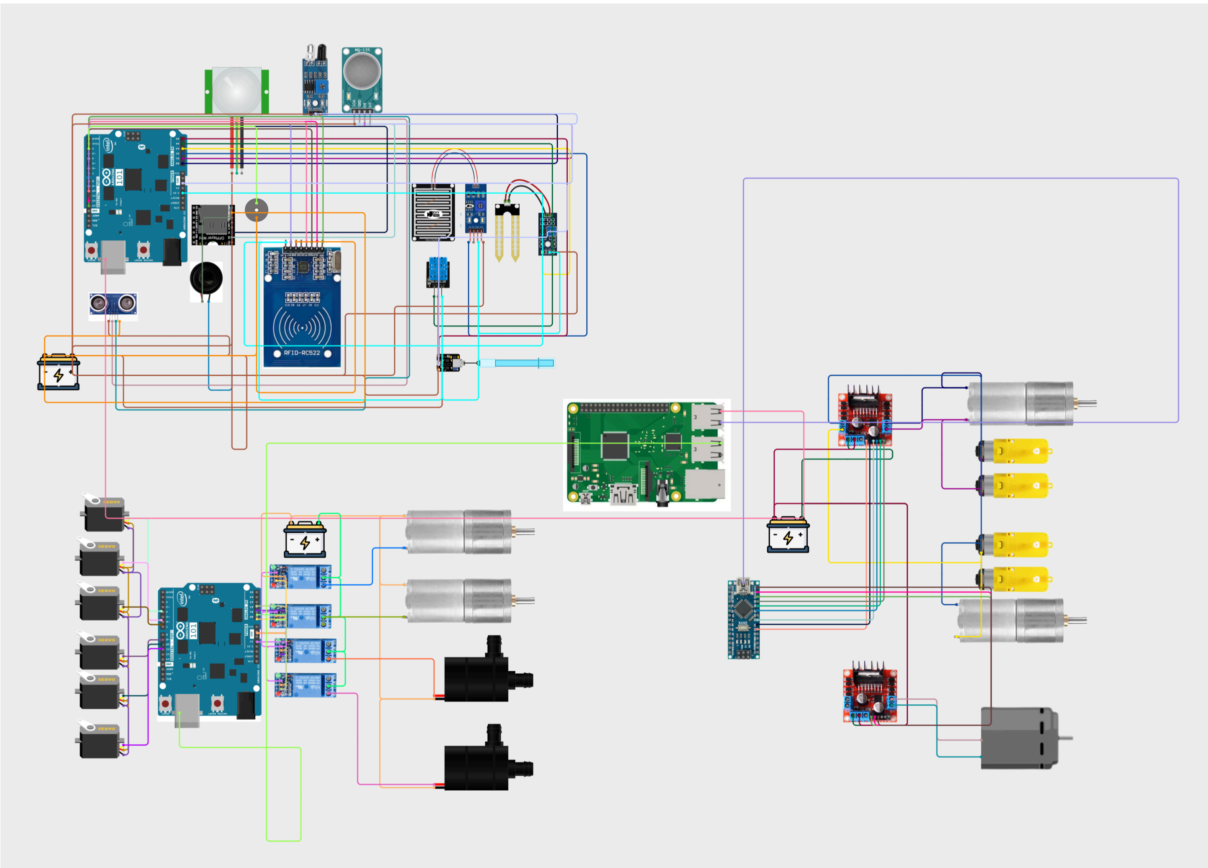

Figure 1: Circuit Diagram of Farm Droid System

This diagram shows the circuit design alongside pin diagram of each components, for easier navigation and reproducibility.

10.2. Data Flow Diagram

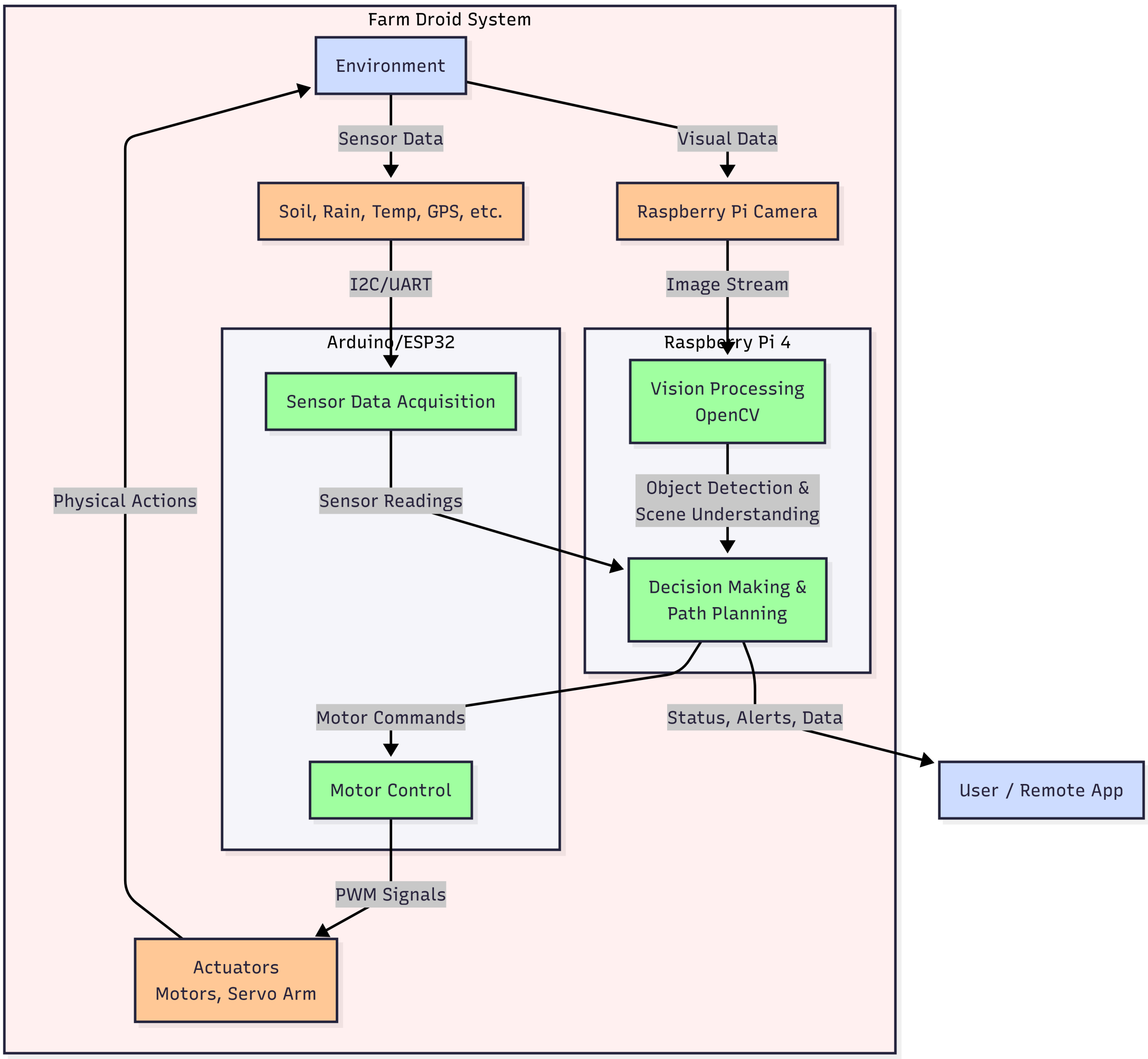

Figure 2: Data flow diagram of Farm Droid system

11. Future Plans

- **Modular Attachments:** Design and develop weeding tool, seed dispenser, watering nozzle etc as modular attachments of the Robotic arm. This will enable the robot to be used in a broader spectrum of agricultural activities.
- **Limited Operations Time:** Due to limited battery capacity, our project can't run continuously for a long time.

12. Conclusion

The Farm Droid is a step towards modernising agriculture by robotics. This project will explore the potential of autonomous mobility and computer vision, to prove farming automation can be agile, intelligent and accessible. The successful completion of this project will result in a working prototype that can be used to move autonomously along paths and carry out basic agricultural activities.

13. References

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